

Certified Defender Move Restriction in Hexo Threat Search

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Abstract

Threat-space search in Hexo has large defender branching because every legal reply must ordinarily be considered. We give a certified restriction based on local defender budgets, cell-ranked obligation guards, and separate completion guards for touched and virgin windows. A verifier accepting these conditions preserves an attacker win over the full legal move set and obtains the resolution bound of the selected certificate path. Further results give an extendable-hit kernel for forced regions, sparse LOSS witnesses, branch-indexed substitution, and finite certificate DAGs. Companion results cover global pairing impossibility, potential-function blocking certificates, local domination patterns, and mechanized evidence for the predecessor search heuristic.

1 Introduction

A df-pn threat-space solver must, at a defender AND node, in principle consider every legal reply. In Hexo this means every empty cell within hex distance 8 of any existing stone, often hundreds of cells. Enumerating these replies dominates the cost of the search.

The restriction studied here is *certified*: every omitted defender reply has a verifier-checkable proof of sound dismissal. Search trimming may cause the solver to miss an attacker win, so it can reduce completeness. It cannot cause a false attacker win to pass verification. This separation between completeness and soundness holds regardless of the heuristic used to generate the searched replies.

The restriction is finite and certificate-relative. Each node carries a local upper bound on the remaining defender placements. Each future attacker placement or leaf-witness empty is a live obligation with its own deadline. Direct obligations, ranked seed guards, touched-window empties, and virgin window seed bands form the certified zone. This local accounting gives the resolution bound of the selected certificate path. No fixed static set is sufficient for every continuation on the unbounded board.

The paper first gives the formal game model in Section 2 and the locality lemmas in Section 3. It then defines zone-carrying certificates in Section 4 and proves pathwise dismissal soundness in Section 5. The constructions of Section 6 give the ranked zone, an extendable-hit kernel, sparse LOSS witnesses, branch-indexed substitution envelopes, and finite DAG certificates. One open problem from the first version, the frontier-band accounting, is resolved by the ranked deadline and touched–virgin exposure analysis. The same section distinguishes the certified zone from a measured predecessor generator whose mean size was 147 cells, compared with 302 legal cells. The remaining layers establish the impossibility of a global pairing defense in Section 7, develop potential-function certificates in Section 8, give local domination patterns in Section 9, and report mechanized evidence in Section 10.

Threat-space search and proof-number methods provide the search setting for these certificates [1, 2]. Pairing strategies and the potential method are standard tools for positional games [3, 4].

The full internal proof documents, review history, and validation scripts are available in the repository [5].

2 The game and formal model

2.1 Informal rules

Hexo is a two-player placement game on an unbounded hexagonal grid. The opening player places one stone at the origin. Thereafter the players alternate turns of two placements. Stones are permanent. A placement is legal on any empty cell within hex distance 8 of a stone of either player. The first player to occupy six consecutive cells on one of the three grid axes wins. A win is checked after each placement, so a win on the first placement ends the game before the second placement of that turn.

We use attacker and defender as fixed roles, denoted by A and D. A ply is one placement, while a normal turn contains two plies. Everything below concerns positions after the Opening placement (the origin is occupied; budgets follow the turn structure of Definition 2.5).

2.2 Formal model

The first definition fixes the infinite board and the metric used by both the line geometry and the legality rule.

Definition 2.1 (Board, cells, and distance). The cells are the integer axial lattice $\mathbb{Z}^2 \ni x = (q, r)$. For $x = (q, r)$ and $y = (q', r')$, set

$$d(x, y) = \max\{|q - q'|, |r - r'|, |(q - q') + (r - r')|\}.$$

The winning patterns are finite segments along the three line directions of the axial lattice.

Definition 2.2 (Axes and windows). The axis vectors are

$$\mathbf{a} \in \{(1, 0), (0, 1), (1, -1)\}.$$

A *window* is a set

$$W = \{s + i\mathbf{a} : 0 \leq i \leq 5\}$$

for a start cell s and an axis vector $\mathbf{a}$. Every window has exactly six cells, and every cell lies in exactly 18 windows, from three axes and six possible offsets on each axis.

The next fact records the metric consequence of belonging to one window.

Fact 2.3 (Window collinearity). Two distinct cells of a window differ by a nonzero multiple of the window's axis vector. Thus any two cells contained in a common window are collinear along that axis and have distance at most 5.

Figure 1 shows the coordinate convention and one distance calculation.

A position records the finite board state and the exact placement remaining in the current turn.

Definition 2.4 (Positions). A position is $P = (\sigma, m, b)$, where σ is a finite partial map $\mathbb{Z}^2 \rightarrow \{A, D\}$, the mover is $m \in \{A, D\}$, and $b \in \{1, 2\}$ is the mover's number of remaining placements in the current turn. Stones are permanent, and $\text{Stones}(P) = \text{dom } \sigma$. For $X \in \{A, D\}$, let $\bar{X}$ denote the other player. For such an X and a window W , define

$$\text{cnt}_X(W, P) = |\{c \in W : \sigma(c) = X\}|.$$

When the position is clear from context, we suppress it and write $\text{cnt}_X(W)$.

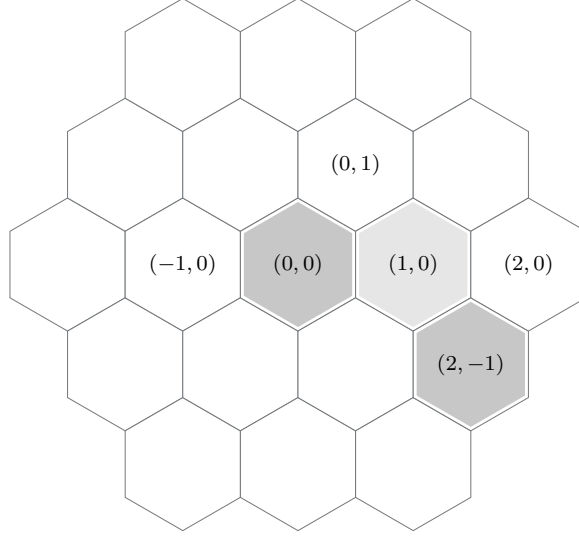
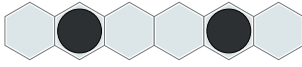
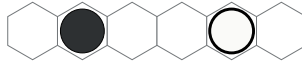


Figure 1: Pointy-top axial coordinates. The shaded cells show a shortest path from $(0, 0)$ to $(2, -1)$, and $d = \max(|2|, |-1|, |1|) = 2$.



(a) Alive for A.



(b) Dead.



(c) An A-threat with count 4.

Figure 2: Alive, dead, and threat window states. Filled disks are attacker stones and open disks are defender stones.

Legality is local to the existing stones, while a transition must also account for immediate termination and the two-placement turn structure.

Definition 2.5 (Legality and transitions). A cell c is *legal* in P if and only if

$$c \notin \text{dom } \sigma \quad \text{and} \quad \exists s \in \text{dom } \sigma : d(c, s) \leq 8.$$

Write $\text{Legal}(P)$ for the set of cells legal in P . The mover places on a legal cell c , producing $\sigma' = \sigma \cup \{c \mapsto m\}$. If some window W then satisfies $\text{cnt}_m(W) = 6$, the game terminates immediately and m wins. Wins are therefore checked after each placement and may occur on the first placement of a two-placement turn. If there is no win, the next position is $(\sigma', m, 1)$ when $b = 2$, and $(\sigma', \bar{m}, 2)$ when $b = 1$.

Window states distinguish usable winning lines, blocked lines, and lines that can be completed within one normal turn.

Definition 2.6 (Window states). A window W is *alive for X* if $\text{cnt}_{\bar{X}}(W) = 0$. An all-empty window is alive for both players. It is *dead* if $\text{cnt}_A(W) \geq 1$ and $\text{cnt}_D(W) \geq 1$. It is an *X-threat* if it is alive for X and $\text{cnt}_X(W) \geq 4$.

The three panels in Fig. 2 give the basic mask patterns used by this terminology.

The implemented one-ply analysis tests whether the mover can win in the remaining placements or whether the opponent's threats exceed the defensive budget.

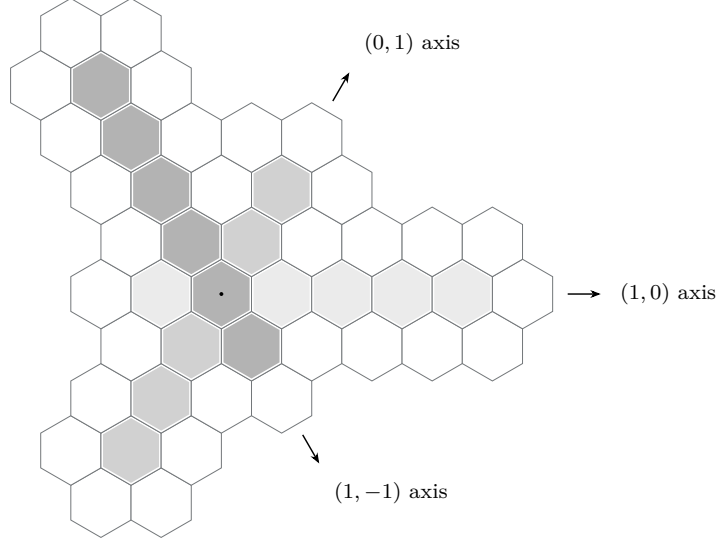


Figure 3: One six-cell window on each axis through the central cell, shown with three fill shades. Sliding a window so that the cell occupies offsets 0 through 5 gives $3 \text{ axes} \times 6 \text{ offsets} = 18$ windows through the marked cell.

Definition 2.7 (λ^1 predicates). For mover m with budget b , $\text{own_win_now}(P)$ holds if and only if there is an m -alive window W such that

$$\text{cnt}_m(W) = 5, \quad \text{or} \quad \text{cnt}_m(W) = 4 \text{ and } b = 2.$$

The value $\text{hit}(P)$ is the multiset of empty-cell sets of the $\bar{m}$ -threat windows. The value $\text{mhs}(P)$ is the minimum number of cells meeting every set in $\text{hit}(P)$, computed exhaustively for values at most 2. The λ^1 verdict is WIN when $\text{own_win_now}(P)$ holds, LOSS when it does not hold and $\text{mhs}(P) > b$, and undecided otherwise. The post-opening soundness of these implemented verdicts is assumed here.

Absolute placement indices allow later certificates to state an exact win horizon without treating a two-placement turn as simultaneous.

Definition 2.8 (Plies and horizon). A *ply* is one placement, and ply indices are absolute along a play. For a position P with mover m , the *defender-placement count before ply T* , written $\mathfrak{D}(P, T)$, is the number of plies strictly before T at which D places, as determined by (m, b) and alternation.

The final convention fixes the direction needed when defender replies are removed from an attacker-win certificate.

Definition 2.9 (Dominance direction). Outcomes are evaluated for the attacker A, with $\text{WIN} \succ \text{nonWIN}$. A *sound dismissal* of a defender reply c at a node N means that, if the certificate proves that A wins against all searched replies, then A also wins against c .

3 Elementary geometry

The first bound relates the number of stones in a window to the maximum gap from an empty cell to those stones.

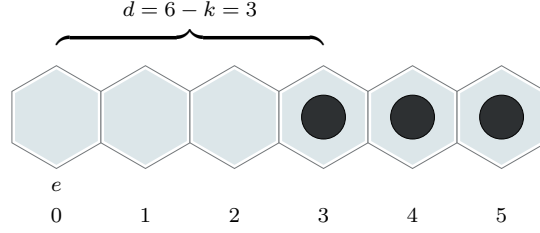


Figure 4: The tightness configuration for the packing bound, shown at $k = 3$. In general the k stones occupy offsets $6 - k$ through 5, while the empty cell at offset 0 is at distance $6 - k$ from the nearest stone.

Lemma 3.1 (Packing bound). *In a window with $k \geq 1$ stones of one colour and no stones of the other, every empty cell of the window is within distance $6 - k$ of a same-window stone. The bound is tight.*

Proof. Index the cells $0, \dots, 5$ along the window axis. Let e be empty, and let δ be its axis-distance from a nearest in-window stone. The δ cells from e toward that stone, including e and excluding the stone, are empty and lie in the window. There are $6 - k$ empty cells in total, so $\delta \leq 6 - k$. By fact 2.3, axis-distance agrees with hex distance for these cells. For tightness, put the stones at offsets $6 - k, \dots, 5$ and take e at offset 0. $\square$

Figure 4 displays the extremal arrangement and a concrete $k = 3$ instance of the general formula.

A mixed set of occupied and empty cells on a finite line segment must change state across some adjacent pair.

Lemma 3.2 (Boundary pair). *Any window containing at least one stone and at least one empty cell has an empty cell at distance 1 from a same-window stone.*

Proof. The six cells are consecutive on a line. The stone cells form a nonempty proper subset of this sequence. At a boundary of that subset, a stone and an empty cell are consecutive. Their hex distance is 1 by fact 2.3. $\square$

The packing bound places all tactics that can resolve within the current turn near stones already on the board.

Theorem 3.3 (Single-turn tactical locality). *Every placement that completes a win within the current turn, and every empty cell of every threat window, lies within distance 2 of an existing stone.*

Proof. Completion within at most two placements requires a window with at least four stones of the mover already present. Apply Lemma 3.1 with $k = 4$ to its empty cells. A threat window also has at least four stones by Definition 2.6, so the same bound applies to every one of its empty cells. $\square$

Creating a threat can start one cell farther away because the placement need only raise an existing three-stone window to count four.

Theorem 3.4 (Threat-creation locality). *A placement creating an X -threat lies within distance 3 of an X stone. The bound is tight.*

Proof. Immediately before the placement, the window that becomes an X -threat held at least three X stones. Apply Lemma 3.1 with $k = 3$ to the placement cell. For tightness, place X stones at offsets 3, 4, 5 of a window and make the threat-creating placement at offset 0. $\square$

4 Zone-carrying certificates

At an AND node, the solver may face hundreds of legal defender replies. A zone-carrying certificate gives each omitted reply a checkable dismissal argument tied to the remaining certificate.

4.1 Transfer lemmas

Every comparison of two positions must account for the same three ways in which a stone can affect the game.

Lemma 4.1 (Difference accounting). *Let $P' = P \pm \{x\}$ differ from P by one stone at the cell x . Every game-relevant predicate difference between P and P' is confined to the following channels:*

- (C1) the occupancy and ownership of x ;
- (C2) the masks of the 18 windows containing x , and hence completion, alive or dead status, threats, `own_win_now`, `hit`, and `mhs`; and
- (C3) the legality frontier $\{c : d(c, x) \leq 8\} \setminus \text{dom } \sigma$.

Proof. The transition rule depends on occupancy at the placed cell, legality, and the masks used for termination. The tactical predicates are functions of window masks, which change only for windows containing x . Legality justifications change through x only within distance 8. There are no other state components. $\square$

Permanence lets later witnesses be interpreted at earlier points of the same play.

Lemma 4.2 (Permanence). *Window masks are monotone nondecreasing along every play. Dead windows remain dead. A window alive for X at ply t is alive for X at every earlier ply.*

Proof. Stones are never removed by Definitions 2.4 and 2.5. $\square$

Adding inert defender stones transfers the cells, masks, legality, and completion events used later in the proof.

Lemma 4.3 (Inert-extension transfer). *Let $P' = P + X$, where X is a finite set of defender stones. Let $\mathcal{W}$ be a set of windows and C a set of cells such that no member of X lies in a member of $\mathcal{W}$ and $X \cap C = \emptyset$. Then:*

- (a) every cell of C empty in P is empty in P' ;
- (b) every $W \in \mathcal{W}$ has identical masks in P and P' ;
- (c) $\text{Legal}(P) \setminus X \subseteq \text{Legal}(P')$, so every cell of C legal in P is legal in P' ; and
- (d) the A-complete windows of P' are exactly the A-complete windows of P disjoint from X .

Thus an A-completion on an X -free window transfers from P to P' , and every A-completion in P' was available in P .

Proof. Parts (a) and (b) follow from occupancy and incident-window masks. Adding stones can only add legality justifications, while occupancy removes the cells of X . An A-complete window contains no defender stone, so windows meeting X are blocked in P' and the other masks agree. $\square$

Deleting defender stones is monotone for attacker tactics, but it requires a separate legality condition for attacker placements.

Lemma 4.4 (Deletion monotonicity and well-formed legality). *Let $P'' = P - Y$, where Y is a finite set of defender stones of P . Then:*

- (a) every A-threat of P is an A-threat of P'' , so $\text{mhs}(P'') \geq \text{mhs}(P)$ for the defender and defender λ^1 -LOSS verdicts transfer from P to P'' ;
- (b) every A-complete window of P is A-complete in P'' ;
- (c) a cell empty in P is empty or newly empty in P'' ; and
- (d) legality may shrink. Every relied-upon attacker placement must be within distance 8 of a stone outside Y . In the uses below it is within distance 3 of an attacker stone, as supplied for threat creation by Theorem 3.4, or within distance 8 of an unaffected root stone.

Proof. Deleting defender stones can only revive attacker windows and cannot remove an attacker completion. Emptiness follows from Definition 2.4. The last condition restores the legality-frontier channel after deletion. $\square$

Completion counting isolates the defender placements relevant to one window.

Lemma 4.5 (Completion counting). *If $\text{cnt}_D(W, P) = s$ and the defender makes at most D placements at plies strictly before T , then W cannot become D-complete before T unless $s + D \geq 6$.*

Proof. A completion needs six defender stones in W . At most s are present in P , and at most D more can be placed before T . $\square$

4.2 Certificate grammar and local clocks

A certificate records the searched tree and checkable witness data for every way in which a selected path can resolve.

Definition 4.6 (Certificate grammar). A *certificate* $\mathcal{C}$ for “A wins from P_0 ” is a finite rooted tree whose nodes have exact positions and mover/budget labels. The root is P_0 and is nonterminal. The verifier recomputes each path-derived ply clock; leaf-ply is the index of the last placement on the path to a leaf. Every edge is the exact legal successor under Definition 2.5, and no edge is defender-terminal. Every maximal node has one of these types.

- An *OR-COMPLETION leaf*, with A to place, names one designated placement, its A-complete witness window, and its completion ply.
- A *WIN leaf*, with A to place, is a verifier-checked `own_win_now` position. It names A-alive count-5 witness windows for either budget, or count-4 witnesses when the budget is 2, and declares resolution within the current attacker turn.
- A *LOSS leaf*, with D to place and budget b , names an A-threat family $\mathcal{T}$ whose empty-set family has transversal number $\tau(\mathcal{T}) > b$, and carries a verifier-checked `¬own_win_now` fact. For every complete nonterminal remainder H of the defender turn, consisting of exactly b placements unless the game ends earlier, some $W_H \in \mathcal{T}$ satisfies $E(W_H, P_L) \cap H = \emptyset$. The attacker then fills its at most two empties. The declared worst-case resolution is `leaf-ply` + $b + 2$. The verifier may require $|\mathcal{T}| \leq 3$ for $b = 1$ and $|\mathcal{T}| \leq 6$ for $b = 2$, by Lemma 6.3.

An internal OR node has one designated attacker placement and its exact child. An internal AND node N has a nonempty searched set $S(N)$ of legal replies, one exact child per searched reply, and a dismissal claim for every other legal reply. The verifier retains the internal defender `¬own_win_now` check as a diagnostic. It is derivable under the completion zone by Lemma 6.10, but remains mandatory at LOSS leaves and an explicit premise of Theorem 6.5. Every resolution is finite, and the global horizon T is the maximum declared resolution over all root-to-leaf paths. Finite consistently labelled DAGs are admitted by Definition 6.11 and Theorem 6.12.

The remaining defender work below a node is measured locally rather than by the maximum resolution elsewhere in the tree.

Definition 4.7 (Admissible local defender budget). The exact local budget $B(N)$, in defender placements, is given by

$$\begin{aligned} B(\text{OR-COMPLETION}) &= B(\text{WIN}) = 0, & B(\text{LOSS}, b) &= b, \\ B(\text{OR}) &= B(\text{child}), & B(\text{AND}) &= 1 + \max_C B(C). \end{aligned}$$

The two placements of a defender turn are two edges, and a LOSS leaf adds its remaining b placements. A verifier may instead accept any nonnegative integral labels satisfying $B(L) \geq b$ at LOSS leaves, $B(N) \geq B(C)$ on OR edges, and $B(N) \geq 1 + B(C)$ on AND edges.

Remark. The first version of this paper required the global horizon count $\mathfrak{D}(P_N, T)$; that requirement was since shown unnecessary.

Only future attacker placements and leaf-entry witness empties must remain unoccupied by dismissed defender stones.

Definition 4.8 (Compressed obligations). Write $E(W, P) = W \setminus \text{Stones}(P)$. A *live obligation role* below N is one of two types:

- (1) a future designated certificate attacker placement, including an OR-COMPLETION placement; or
 - (2) a triple (L, W, y) for a named WIN or LOSS witness W at a leaf L and a cell $y \in E(W, P_L)$.
- Witness-empty roles represent every possible leaf continuation placement, including the adaptive LOSS choice. If $\Omega(N)$ is the set of all such roles at reachable descendants, define

$$\text{Obl}(\mathcal{C}, N) = \text{Prot}(N) := \left\{ y : \begin{array}{l} \text{some live role in } \Omega(N) \\ \text{is carried by } y \end{array} \right\}.$$

An attacker-placement role is live through its check immediately before the OR step. A witness-empty role is live through its check at leaf entry. Each is discharged immediately afterward. Reachable-descendant nesting gives $\text{Prot}(M) \subseteq \text{Prot}(N)$ whenever M descends from N .

Remark. The first version of this paper protected every cell of each witness window; that requirement was since shown unnecessary.

Different cells can be needed at different depths, so each role has its own defender-placement deadline.

Definition 4.9 (Cell-specific deadlines). The deadline of a role ρ is its designated attacker placement or, for a WIN or LOSS witness-empty role, leaf entry. At a node N from which ρ is reachable, $r_N(\rho)$ is an integral upper bound on defender placements before that deadline. Exact ranks count the maximum number of AND edges on a reachable path. The verifier checks $r = 0$ at the deadline, $r_N(\rho) \geq r_C(\rho)$ across an OR edge, and $r_N(\rho) \geq 1 + r_C(\rho)$ across an AND edge while the role is live. For a cell carrying several roles, put

$$r_N(y) = \max\{r_N(\rho) : \rho \in \Omega(N) \text{ is carried by } y\}.$$

A role is discharged only after its $r = 0$ deadline check. A seed band is formed only at an internal AND node with $r_N(y) \geq 1$. LOSS witness empties are protected only through leaf entry; their adaptive contract controls the remaining b placements. Defender-completion windows use Definitions 4.7 and 4.10, not role ranks. The uniform admissible choice is $r_N(\rho) = B(N)$ before the deadline.

Completion risk also has a separate local clock for every window.

Definition 4.10 (Per-window defender exposure). For each window W , let $E_Q^D(W)$ be the maximum number of future defender placements before the certificate attacker wins or first places in W . Its exact recurrence is

$$E_Q^D(W) = \begin{cases} 0, & Q \text{ is WIN or OR-COMPLETION,} \\ b, & Q \text{ is LOSS with defender budget } b, \\ 0, & Q \text{ is an OR whose move enters } W, \\ E_C^D(W), & Q \text{ is any other ordinary OR,} \\ 1 + \max_C E_C^D(W), & Q \text{ is an internal AND.} \end{cases}$$

If W is already not alive for D, set $E_Q^D(W) = 0$. An attacker move *enters* W when W becomes permanently not alive for D; it need not become dead. Overlapping windows have separate clocks. The stopping attacker placement is also a live obligation, so the obligation zone keeps the stop playable in the real game.

The local labels decrease along every selected path and preserve the relevant stone-count bounds.

Lemma 4.11 (Local-clock and nesting facts). *If a descendant M is reached from N after k defender placements, then*

$$\begin{aligned} B(M) + k &\leq B(N), \\ \text{cnt}_D(W, P_M) + B(M) &\leq \text{cnt}_D(W, P_N) + B(N). \end{aligned}$$

Every selected path below N , including a LOSS remainder, contains at most $B(N)$ defender placements before its declared attacker resolution. For every still-live role ρ , $r_M(\rho) + k \leq r_N(\rho)$. Before the attacker enters W ,

$$\begin{aligned} E_M^D(W) + k &\leq E_N^D(W), \\ \text{cnt}_D(W, P_M) + E_M^D(W) &\leq \text{cnt}_D(W, P_N) + E_N^D(W). \end{aligned}$$

Also $E_N^D(W) \leq B(N)$, every exact rank $r_N^(\rho) \leq B(N)$, and $\text{Prot}(M) \subseteq \text{Prot}(N)$.*

Proof. Sum the verifier inequalities along the path. Each AND edge contributes one and each OR edge contributes zero; the AND maximum covers the selected child, and the LOSS bound covers its remainder. A window gains at most one defender stone per counted defender edge. The exposure recurrence gives the analogous inequalities until its stop and is bounded pointwise by the budget recurrence. An exact role deadline occurs no later than the path resolution. The last claim is reachable-descendant inclusion. $\square$

The zone separates obligation occupancy from the two forms of completion risk. Touched windows need no frontier term; virgin windows need a seed band.

Definition 4.12 (Zone-carrying certificate). For $d(c, W) = \min_{w \in W} d(c, w)$ and $B_r(U) = \{c : \exists u \in U, d(c, u) \leq r\}$, define at each internal AND node N

$$\begin{aligned} Z_{\text{dir}}(N) &= \text{Prot}(N) \cap \text{Legal}(P_N), \\ Z_{\text{seed}}(N) &= \bigcup_{\substack{y \in \text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N)) \\ r_N(y) \geq 1}} \left(\text{Legal}(P_N) \cap B_{8(r_N(y)-1)}(\{y\}) \right), \\ Z_{\text{touch}}(N) &= \bigcup \left\{ E(W, P_N) : \begin{array}{l} W \text{ is alive for D, } \text{cnt}_D(W, P_N) \geq 1, \\ \text{cnt}_D(W, P_N) + E_N^D(W) \geq 6 \end{array} \right\}, \\ Z_{\text{virgin}}(N) &= \left\{ c \in \text{Legal}(P_N) : \begin{array}{l} \text{some all-empty window } W \text{ has } E_N^D(W) \geq 6, \\ d(c, W) \leq 8(E_N^D(W) - 6) \end{array} \right\}. \end{aligned}$$

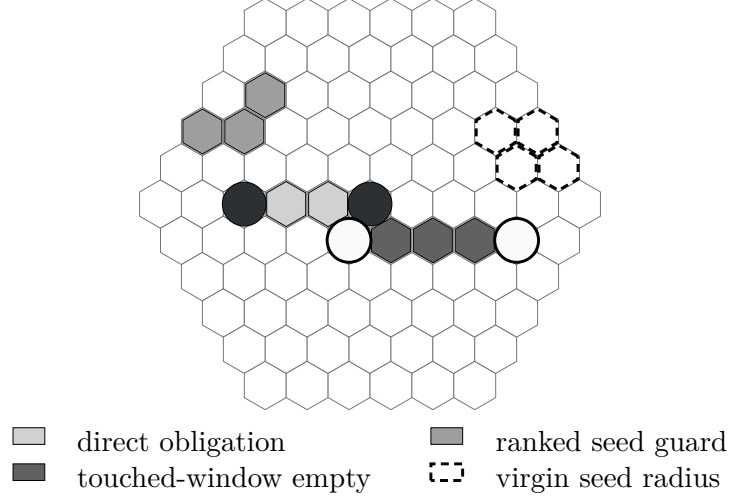


Figure 5: Schematic ranked zone. Direct obligation cells form Z_{dir} . A currently illegal obligation uses radius $8(r_N(y) - 1)$, uniformly $8(B(N) - 1)$. Touched completion windows contribute their empties when $\text{cnt}_D + E_N^D \geq 6$. An all-empty window with exposure at least 6 contributes legal seeds within radius $8(E_N^D - 6)$. Filled disks are attacker stones and open disks are defender stones.

Every cell of Z_{touch} is legal by Lemma 3.1. The certificate is *zone-carrying* when $S(N)$ is independently nonempty at every internal AND node and the following clauses hold.

- (Z2) $S(N) \supseteq Z_{\text{dir}}(N) \cup Z_{\text{touch}}(N) \cup Z_{\text{virgin}}(N)$.
- (Z4) Every certificate attacker placement, including a placement allowed by a leaf contract, lies within distance 8 of an attacker stone in its predecessor position or of a root stone unaffected by the coupling.
- (Z5') $S(N) \supseteq Z_{\text{seed}}(N)$.

With the uniform choice $r_N(y) = B(N)$, clause (Z5') becomes

$$S(N) \supseteq \text{Legal}(P_N) \cap B_{8(B(N)-1)}(\text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N))).$$

All four components are finite and verifier-enumerable. The virgin query is inverted from each legal candidate over the finite radius allowed by $E_N^D(W) \leq B(N)$.

Remark. The first version of this paper required current threat-hitting cells as clause (Z1); that requirement was since shown unnecessary.

The ranked seed guard prevents the first real-only occupation of a protected cell, which is the property used by the coupling.

Lemma 4.13 (First protected occupation). *Suppose direct legal protected cells are searched and every ghost-legal dismissal x at N lies outside radius $8(B(N) - 1)$ of*

$$\text{Prot}(N) \setminus (\text{Legal}(P_N) \cup \text{Stones}(P_N)).$$

Then no defender placement creates a real-only stone in the current protected set. In the ranked form, replace $B(N)$ for each target y by its maximum live role rank $r_N(y)$; the conclusion holds through that role's deadline.

Proof. Assume that the first violation is a defender placement at y . If y is ghost-legal, then $y \in Z_{\text{dir}}$ and is searched, so it is placed in both games. Otherwise trace its real-only legality witness backward

through real-only defender stones to the first ghost-legal dismissed seed x_0 . If y is the p th defender placement in this chain, counting x_0 , there are at most $p - 1$ radius-8 links and

$$d(x_0, y) \leq 8(p - 1).$$

Protection nesting puts $y \in \text{Prot}(N_{x_0})$. At the violation it is still ghost-empty and ghost-illegal. Ghost legality of an unoccupied cell is monotone, and permanence makes it a non-stone at N_{x_0} . The local budget gives $p \leq B(N_{x_0})$; in the ranked form, the still-live role gives $p \leq r_{N_{x_0}}(y)$. Thus x_0 lay in the applicable seed band, contrary to its dismissal. $\square$

The uniform radius is sharp: a ghost-legal seed can be followed by $B - 1$ successive distance-8 placements, with the protected target last.

Remark. The first version of this paper required an $8\mathfrak{D}(P_N, T)$ frontier band; that requirement was since shown unnecessary.

The proof compares the actual play with a certificate walk using canonical defender-stone differences.

Definition 4.14 (Real-ghost coupling). The real position R and ghost position G have equal ply, mover, and budget, and identical attacker stones. Define

$$X = \text{Stones}_D(R) \setminus \text{Stones}_D(G), \quad Y = \text{Stones}_D(G) \setminus \text{Stones}_D(R).$$

The maintained invariants are: every $x \in X$ entered at a dismissal whose ordinary zone test or selected Definition 6.8 envelope certified it outside the live obligations; every $y \in Y$ entered as a searched ghost filler; and $X \cap \text{Prot}(N) = \emptyset$ at each visited node. The previous lemma supplies the last invariant for ghost-illegal dismissals, and obligation nesting preserves it after descent. A history $\hat{X} \supseteq X$ contains every cell ever added to X and never deletes cells. For every window W ,

$$\text{cnt}_D(W, R) = \text{cnt}_D(W, G) + |X \cap W| - |Y \cap W| \leq \text{cnt}_D(W, G) + |\hat{X} \cap W|. \quad (\text{MI})$$

The touched-virgin split excludes every real defender completion before the mapped attacker stop, including mixed legality chains.

Lemma 4.15 (Split completion safety). *In the coupling of Definition 4.14, under Items (Z2) and (Z4) and clause (Z5'), no real defender play completes a fixed window W before the mapped certificate attacker wins or first enters W .*

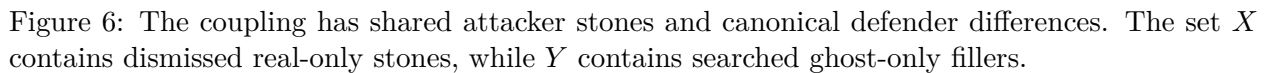
Proof. Suppose W is first completed in the real game before that stop. Shared attacker stones and permanence make it alive for D throughout the relevant ghost prefix. If no W -cell was ever dismissed, then $\hat{X} \cap W = \emptyset$ and (MI) forces the ghost line to have at least the real defender count. This contradicts the exact successors and the ban on defender-terminal edges.

Otherwise let $x \in W$ be the first ever-dismissed real-only fill, at N . If the ghost already has a defender stone in W , every empty of W is ghost-legal. Before x , (MI) bounds the real count by the ghost count. The real placements from x through completion are covered by $E_N^D(W)$, so

$$\text{cnt}_D(W, P_N) + E_N^D(W) \geq 6.$$

The cell x lies in Z_{touch} and should have been searched.

Suppose instead that the ghost window is virgin. If x is ghost-legal, all six fills remain in the exposure count. Hence $E_N^D(W) \geq 6$ and $d(x, W) = 0$, so $x \in Z_{\text{virgin}}$. If x is ghost-illegal, trace its


$$d(x_0, W) \leq 8j, \quad E_{N_{x_0}}^{\mathbf{D}}(W) \geq j + 6.$$

At a LOSS remainder, the coupling stops at leaf entry. If W has an ever-dismissed fill, the same count remains valid because the exposure includes the leaf's b placements. If it has none, (MI) at the leaf bounds the real count by the ghost count. The mandatory leaf `-own_win_now` check bounds the latter by 3 at $b = 2$ and 4 at $b = 1$, so the remaining b placements reach at most 5. $\square$

5 Pathwise dismissal soundness

The main result transfers the selected certificate path to every legal real defender play.

Theorem 5.1 (Local-clock, pathwise dismissal soundness). Let $\mathcal{C}$ obey Definitions 4.6 to 4.10, 4.12 and 4.14 and be zone-carrying under Items (Z2) and (Z4) and clause (Z5'). For every real defender play against the compiled attacker strategy, either the real attacker wins strictly earlier than the mapped ghost resolution, or the play maps to a finite certificate path and A completes by that path's declared resolution. Consequently A wins by the global maximum T , and every certified dismissal is sound.

Proof. Couple the real position R , initially P_0 , to a ghost walk G on $\mathcal{C}$, maintaining Definition 4.14. Process a typed leaf as soon as it is reached.

Step O (OR node, designated placement c). The live placement role puts c in Prot at every current ancestor, so $c \notin X$. Ghost emptiness gives $c \notin Y$. Thus c is real-empty. Clause Item (Z4) supplies a shared attacker or root legality witness. Play c in both games.

At an OR-COMPLETION, the other five cells of the named window are shared attacker stones. The ghost window contains no defender stone, hence no Y -cell, and c is not an X -cell. The real placement completes the same window at the declared ply. If the real placement instead completes another window that the ghost does not, Lemmas 4.3 and 4.4 give an immediate, strictly earlier real attacker win. Otherwise discharge the role after its deadline check and descend to the exact child.

Step A (AND node N , real defender placement d). Exactly one of three cases holds: d is ghost-occupied, ghost-empty and searched, or ghost-empty and unsearched.

Filler subroutine. When the ghost must consume a defender ply without copying d , choose any $d_0 \in S(N)$. Independent nonemptiness is part of the certificate checks. The cell is ghost-empty and ghost-legal. If $d_0 \in X$, the ghost placement removes it from X ; otherwise it adds d_0 to Y . Recompute the canonical differences. A new Y -cell is a searched filler and X only shrinks.

A1. If d is ghost-empty and searched, both games play d and descend to its exact child. The difference sets do not change.

A2. If d is ghost-occupied, then $d \in Y$. The real placement removes it from Y . Run the filler subroutine and descend to the filler child.

A3. Suppose d is ghost-empty and unsearched. If it is ghost-legal and protected, then $d \in Z_{\text{dir}} \subseteq S(N)$, a contradiction. A ghost-legal dismissal is therefore outside Prot and, because it is outside Z_{seed} , satisfies Lemma 4.13. If d is ghost-illegal, that lemma says its placement cannot create the first real-only protected stone. It is again outside the current Prot. The real game plays d and adds it to X and $\hat{X}$. Run the filler subroutine and descend. First protected occupation and obligation nesting preserve $X \cap \text{Prot} = \emptyset$. This accounts for occupancy and the legality frontier. Split completion safety, Lemma 4.15, excludes a real defender completion before the mapped attacker stop.

WIN leaf. Let W be a named attacker-alive witness. Its ghost attacker stones are shared. Every other cell is in $E(W, P_L)$ and has a live witness-empty role, so it is X -free at leaf entry. Attacker-aliveness excludes ghost defender stones and hence Y -cells. The real and ghost masks and empty sets agree. By Lemma 3.1, the one or two empties lie within distance 2 of shared attacker stones. They are real-legal and remain empty. The attacker fills them with no intervening defender placement and completes by the declared resolution.

LOSS leaf. The same compressed-obligation argument makes the real and ghost masks and empty sets identical for every $W \in \mathcal{T}$ at leaf entry. Thus the real transversal number is the checked value $\tau(\mathcal{T}) > b$.

A real defender `own_win_now` is impossible. If its witness window has no ever-dismissed fill, (MI), the shared attacker mask, and the equal budget transfer the predicate to the ghost, contrary to the mandatory leaf check. If it has an ever-dismissed fill, the first-dismissal exposure argument of Lemma 4.15 includes the LOSS leaf's b placements and precludes completion within the current defender turn.

Discharge the leaf roles and let the real defender choose any complete nonterminal remainder H . Since $|H| \leq b < \tau(\mathcal{T})$, some $W_H \in \mathcal{T}$ has $E(W_H, P_L) \cap H = \emptyset$. It remains alive for A. Its at most

two surviving empties lie within distance 2 of shared, permanent attacker stones, so the attacker completes it in the following turn by leaf-ply + $b + 2$. No protection is needed after leaf entry; the adaptive quantifier over H supplies the surviving witness.

Every nonterminal step descends the finite certificate. In the absence of an earlier real attacker win, the coupled play reaches one of these terminal transfers and resolves by that selected path's declared ply. The global T is their maximum. $\square$

6 Computable relevance zones

The soundness theorem gives an explicit ranked zone at each AND node. Its four terms use separate clocks for obligation occupancy and defender completion.

Theorem 6.1 (Ranked zone reading). At an ordinary AND node, the verifier zone is

$$Z(N) = Z_{\text{dir}}(N) \cup Z_{\text{seed}}(N) \cup Z_{\text{touch}}(N) \cup Z_{\text{virgin}}(N).$$

Any independently nonempty searched set $S(N) \supseteq Z(N)$, together with [Item \(Z4\)](#), is sufficient. If $Z(N)$ is empty, one arbitrary legal reply must still be searched. The uniform admissible deadline $r_N(y) = B(N)$ gives the radius- $8(B(N) - 1)$ seed band in [Definition 4.12](#). Exact cell ranks can only reduce that band, and exact window exposures can reduce the corresponding B -clock completion guards. Larger admissible upper bounds remain sound. Current threat-hitting cells are optional search candidates. They enter only the separate kernel of [Theorem 6.5](#).

Proof. The four terms and independent fallback are exactly the requirements of [Definition 4.12](#). Apply [Theorem 5.1](#). $\square$

For a short threat-creating continuation, early attacker obligations remain inside windows already touched by current attacker stones.

Lemma 6.2 (Short-range attacker-obligation containment). *Suppose all attacker placements in a certificate are threat-creating. If a placement is the attacker's k th future placement from N , then for $k \leq 3$ its window contains at least*

$$3 - (k - 1) \geq 1$$

attacker stones already present in P_N . Its obligation role therefore lies in an attacker-touched window.

Proof. Immediately before the placement, its window has at least three attacker stones. At most $k - 1$ are earlier future placements, leaving at least $3 - (k - 1)$ current stones. $\square$

From the fourth future placement onward, a setup cell can lie in a currently virgin window and must be supplied by the obligation set.

LOSS leaves need only a small subfamily of their available threat witnesses.

Lemma 6.3 (Sparse LOSS witnesses). *A LOSS witness family may be chosen with $|\mathcal{T}| \leq 3$ when $b = 1$ and $|\mathcal{T}| \leq 6$ when $b = 2$, where b counts placements.*

Proof. Every threat empty-set has size one or two. For $b = 1$, if the family contains $\{a\}$, select it and one set missing a . Otherwise select $E = \{a, b\}$, one set missing a , and one missing b . No point meets all selected sets. The three edges of a triangle show sharpness.

For $b = 2$, take a maximal pairwise-disjoint subfamily. It cannot have size one, since its sole set would meet the full family. If it has size at least three, three disjoint sets suffice. Otherwise take two disjoint sets E_1, E_2 . Their two-point transversals are the at most four cross-pairs choosing one point from each. For every cross-pair, select one original set that it misses. Together with E_1, E_2 , at most six sets exclude every two-point transversal. The six edges of K_4 show that this general rank-two bound is sharp. Since a LOSS remainder has $|H| \leq b$, a selected family with transversal number greater than b retains a witness disjoint from H . $\square$

A common radius-three generator covers the full zone only under joint local budget and witness-location hypotheses.

Theorem 6.4 (Static-set coverage for short local budgets). Suppose $B(N) \leq 3$, there are at most three remaining attacker placements and all are threat-creating, every named witness-empty role lies in an attacker-touched alive window, and $Z_{\text{seed}}(N)$ is empty. Then

$$r_3(P_N) \cup \{\text{empties of attacker-touched alive windows at } P_N\},$$

where

$$r_3(P) = \{c \in \text{Legal}(P) : \exists s \in \text{Stones}(P), d(c, s) \leq 3\},$$

covers $Z(N)$. Without all the joint hypotheses, this static set is a heuristic candidate generator rather than a certified zone.

Proof. Since $E_N^D(W) \leq B(N) \leq 3$, the virgin term is empty. A touched window in Z_{touch} has at least $6 - E_N^D(W) \geq 3$ defender stones, so all its empties lie within distance 3 of a defender stone by Lemma 3.1. The previous lemma covers the stated attacker-placement roles, the witness hypothesis covers the remaining direct roles, and the seed term is empty. $\square$

In a fully forced region, a residual-transversal kernel can replace the ranked zone while preserving the same path bound.

For the current attacker-threat empty-set family $\mathcal{F}_N$, write

$$\mathcal{F}_N \setminus d = \{E \in \mathcal{F}_N : d \notin E\}, \quad K_b(N) = \{d \in \text{Legal}(P_N) : \tau(\mathcal{F}_N \setminus d) \leq b - 1\}.$$

Theorem 6.5 (Extendable-hit kernel). A kernel-governed region is entered at a certificate node where the real and ghost positions are equal. It uses the following rule until a typed terminal or a handoff with $\text{mhs} > b$. At every internal AND node in the region, require the explicit defender $\neg \text{own_win_now}$ check and

$$\text{mhs}(P_N) = \tau(\mathcal{F}_N) \leq b,$$

and search exactly $K_b(N)$. No original obligation or core term is required. If $\text{mhs} < b$, then $K_b(N) = \text{Legal}(P_N)$ and there is no pruning. If $\text{mhs} = b$, the kernel weakly refines the current hitting-cell set and can strictly reduce it; equality can occur. A node with $\text{mhs} > b$ ends the kernel region. It must be a valid LOSS leaf or hand off to an existing nonempty sound subtree. The core-free claim does not apply below that handoff without a fresh equal-position entry.

Proof. If $\tau < b$, deleting the sets hit by any legal d cannot increase τ , so every legal reply lies in K_b . If $\tau = b$, follow a minimum transversal. Its first cell is legal by Theorem 3.3 and lies in K_b , so the kernel is nonempty.

Before the first real reply $d \notin K_b$, the real and certificate plays agree and $X = Y = \emptyset$. Abandon the original subtree and use the residual current threat family. For $b = 2$, the successor has defender budget 1 and residual transversal number greater than 1, giving an adaptive LOSS refutation. For

$b = 1$, a current attacker threat survives and the attacker has budget 2, giving a WIN refutation. No original future obligation is used.

The defender cannot complete first. The explicit immediate-win premise is false. It bounds every defender-alive window by count 3 at $b = 2$ and count 4 at $b = 1$. The remaining two or one defender placements reach at most 5.

For the same- T claim, follow a minimum hitting line through the kernel. At $b = 2$, its first member leaves a family hit by the second, and the kernel-governed $b = 1$ child searches that second member. At $b = 1$, the sole common hit is searched. These placements kill every current attacker threat, and defender stones create no new attacker threat. The first following attacker placement cannot complete a window: such a window would have been a current count-5 threat and was killed. The corresponding original path therefore reaches the second attacker placement required by the auxiliary refutation.

Finally, if $\tau > b$ and $d \in K_b$, then d together with a residual transversal of size at most $b - 1$ would meet all of $\mathcal{F}_N$, a contradiction. Thus K_b is empty outside the stated scope and would violate the independent nonemptiness rule. $\square$

6.1 Ranked relevance closure

The mandatory closure is the ranked zone itself. Search may add local heuristics without making them verifier premises.

Definition 6.6 (Ranked relevance closure). Define

$$R_{\text{cert}}(\mathcal{C}, N) = Z_{\text{dir}}(N) \cup Z_{\text{seed}}(N) \cup Z_{\text{touch}}(N) \cup Z_{\text{virgin}}(N).$$

For search, the solver may use the finite superset

$$R_{\text{search}}(\mathcal{C}, N) = R_{\text{cert}}(\mathcal{C}, N) \cup \text{hitting}(P_N) \cup \mathcal{A}(P_N) \cup r_3(P_N),$$

where $\text{hitting}(P)$ is the union of the current attacker-threat empty sets, $\mathcal{A}(P)$ is the set of empties in attacker-touched alive windows, and $r_3(P)$ is the legal radius-3 neighborhood of current stones. The last three terms are heuristics, not hypotheses of Theorem 5.1. Obligations are finite because the certificate is finite, and touched windows are finite because the position is finite. For each finite legal candidate, the virgin test examines only windows within its bounded exposure radius.

The verifier must reconstruct every role and clock used by the mandatory closure.

Theorem 6.7 (Ranked-closure coverage). At every ordinary zone-governed AND node, any independently nonempty $S(N) \supseteq R_{\text{cert}}(\mathcal{C}, N)$ satisfies Item (Z2) and clause (Z5'). Dismissal soundness follows from Theorem 5.1 when the certificate grammar, the budget, rank, and exposure checks, and Item (Z4) also hold. In constructing R_{cert} , the verifier unions roles over all reachable descendants, includes every designated attacker move including OR-COMPLETION, treats WIN and LOSS continuation cells as leaf-entry witness-empty roles, takes the maximum rank over live roles sharing a cell, and forms no band after the $r = 0$ deadline check. Defender-completion windows remain on the E^D clock or the conservative B clock.

Proof. These requirements are exactly the four components and three live verifier labels of Definition 4.12. Apply Theorems 5.1 and 6.1. $\square$

6.2 Branch-indexed substitution

A dismissal may select one searched child and verify a smaller envelope over that child's reachable descendants. The current transition must be included in every clock.

Definition 6.8 (Transition-inclusive substitution envelope). At an AND node N , annotate each ghost-legal dismissal d with a searched substitute

$$\varphi_N(d) = s \in S(N)$$

whose exact child is C_s . Declare a nonempty fallback $F(N) \subseteq S(N)$ independently of these annotations; in particular, $S(N)$ is not defined as only the replies with no safe substitute. A valid selected envelope has the following data and checks.

- (1) Its transition budget is $\widehat{B}(N, d, s) = 1 + B(C_s)$, including the current real/ghost defender ply and every reachable LOSS remainder.
- (2) Its obligations are the union of roles over all reachable descendants of C_s , not one leaf or selected continuation.
- (3) The real cell d avoids those obligation cells and every transition-dangerous completion empty.
- (4) For a child obligation y with rank $r_{C_s}(y)$ and $y \notin \text{Legal}(P_N) \cup \text{Stones}(P_N)$, direct occupation $d = y$ is forbidden and the parent seed test is

$$d(d, y) > 8r_{C_s}(y) = 8((1 + r_{C_s}(y)) - 1).$$

- (5) For each defender-alive touched window W with $\text{cnt}_D(W, P_N) \geq 1$, the $\widehat{B}$ form forbids $d \in E(W, P_N)$ when $\text{cnt}_D(W, P_N) + 1 + B(C_s) \geq 6$. For an all-empty W , it forbids a seed with $d(d, W) \leq 8(\widehat{B} - 6)$ when $\widehat{B} \geq 6$. The exact exposure form instead sets $\widehat{E}^D(W) = 1 + E_{C_s}^D(W)$, uses $\text{cnt}_D(W, P_N) + \widehat{E}^D(W) \geq 6$ for a touched fill, and uses $d(d, W) \leq 8(\widehat{E}^D(W) - 6)$ with $\widehat{E}^D(W) \geq 6$ for a virgin window. Both touched and virgin clauses are mandatory for the selected clock.
- (6) The fallback $F(N)$, and hence $S(N)$, remains nonempty independently of whether any dismissal has a safe substitute.
- (7) For a ghost-legal A3 dismissal, the ghost plays $\varphi_N(d)$. Case A2 may use any searched filler. A ghost-illegal A3 dismissal inherits the envelope of the earlier ghost-legal seed supporting its real legality and may use any searched filler in the current reachable subtree; it does not start a child-only clock.
- (8) The envelope protects every reachable LOSS witness-empty role through leaf entry, while $\widehat{B}$ or $\widehat{E}^D$ counts the leaf's b placements for defender completion and defender-own-win exclusion. The default-child form declares $f(N) \in F(N)$ and uses $C_{f(N)}$ for every ghost-legal dismissal. It must satisfy the same transition-inclusive tests; a child obligation union alone is insufficient.

These envelopes preserve the coupling because their reachable sets remain nested after the substitution.

Theorem 6.9 (Branch-indexed substitution soundness). The conclusion of Theorem 5.1 remains valid if an AND node's global dismissal tests are replaced, dismissal by dismissal, by valid Definition 6.8 envelopes. The same holds for the default-child form.

Proof. The real placement d and ghost substitute s consume the same defender ply, so clock, mover, and budget remain synchronized. The avoidance check supplies occupancy safety. The seed and completion checks supply legality- frontier and window-mask safety with the current transition

included. Thereafter all ghost nodes are reachable descendants of C_s . Case A1 may take any searched edge there; A2 adds no X -stone and may take any filler. At a later ghost-legal A3 step, the selected descendant set nests inside the earlier one. A ghost-illegal A3 step traces its legality chain to an earlier tested seed and inherits that envelope. Every earlier X -stone therefore continues to avoid current roles, Lemma 4.13 applies with the transition rank, and Lemma 4.15 applies with $\widehat{B}$ or $\widehat{E}^D$. The LOSS check covers the post-leaf portion. Attacker legality, compressed witness transfer, and all terminal cases of Theorem 5.1 are unchanged. The real play either wins earlier or resolves on the selected reachable path. $\square$

The added transition unit is necessary in both channels. For the legality frontier, let N have $b = 2$, let legal d be at distance 8 from a ghost-illegal future obligation y , and let C_s have one defender placement before a shared legal attacker setup makes designated y legal. Then $r_{C_s}(y) = 1$. A child-only radius $8(r - 1) = 0$ permits d ; the real move legalizes y and the remaining defender placement occupies it. The parent radius $8r = 8$ forbids d . For completion, let a defender-alive window have two defender stones at N , with $d \in W$, $s \notin W$, and $B(C_s) = 3$. The child-only count is $2 + 3 = 5$, but real d and three later fills complete W . The transition-inclusive count is $2 + 1 + 3 = 6$ and forbids the dismissal.

The ordinary completion zone also explains why the internal immediate-win check is diagnostic rather than an extra hypothesis.

Lemma 6.10 (Internal own-win diagnostic). *Under the completion-zone requirement of Definition 4.12 and the ban in Definition 4.6 on defender-terminal edges, an internal AND node cannot have a defender-alive count-5 window, or a count-4 window when $b = 2$.*

Proof. A count-5 touched window has $E_N^D(W) \geq 1$, so its legal last empty lies in Z_{touch} and must be searched. Its exact child is defender-terminal, which is forbidden. At $b = 2$, a count-4 touched window has exposure at least two before the attacker can move, so both empties are searched. After either first fill, the $b = 1$ child has count 5. It cannot be a LOSS leaf because the mandatory leaf check fails; if it is internal, its last empty is searched and gives the same forbidden terminal edge. $\square$

The diagnostic remains mandatory at LOSS leaves and an explicit premise of the kernel theorem, whose region has no completion guard.

6.3 Finite certificate DAGs

Shared certificate subtrees are sound when all shared labels and clocks are exactly consistent.

Definition 6.11 (Finite acyclic certificate DAG). A certificate may be a finite rooted acyclic graph if every shared node has one exact Definition 4.6 label: the same position, mover and budget, designated OR action or AND searched-successor map, leaf witness data, and one consistent path clock. Obligations at a DAG node are unions over all reachable descendants. Local budgets, ranks, and exposures satisfy their inequalities on every outgoing edge. Coupling histories X , Y , and $\widehat{X}$ remain path-local.

Finite unfolding reduces the DAG form to the tree theorem.

Theorem 6.12 (DAG unfolding). Every Definition 6.11 DAG satisfying the ordinary Definition 4.12 rules or the substitution rules of Definition 6.8 has the corresponding conclusion of Theorem 5.1.

Proof. A finite acyclic graph has finitely many root-to-node paths. Unfold one copy of each shared node along each such path. Exact labels and consistent clocks give a finite Definition 4.6 tree with the same legal transitions, terminal data, and local-clock inequalities. Reachability remains nested along every edge, so descendant obligation unions and protection monotonicity are preserved. Apply Theorem 5.1 or Theorem 6.9. The unfolded coupling histories are exactly the required path-local histories. $\square$

The solver may search with R_{search} , a smaller heuristic, or a substitution-aware generator. Verification must establish ordinary R_{cert} coverage and attacker legality, or validate every substitution annotation, while retaining an independently nonempty fallback. A failure yields UNKNOWN. Search trimming affects completeness only.

The predecessor heuristic measured a mean of 147 cells against 302 legal cells on random midgame positions. These figures describe the conservative search generator, not the ranked certificate-relative verifier zone.

7 Impossibility of global pairing

A pairing strategy preassigns disjoint pairs and answers each attacker stone in a paired cell by taking its mate. In a single-placement game this gives a perfect local defense. Under Hexo’s two-placement turns, the attacker may occupy both cells of a pair within one turn, so even an existing pairing would need a separate argument. The theorem below makes the question moot: the required matching does not exist.

Theorem 7.1 (No pairing; source T8). *There is no partial matching M on the cells, with each cell in at most one pair, such that every window contains both cells of some pair.*

Proof. Discard every pair that is not contained in a window. This removes every non-axis-aligned pair and every axis pair at distance greater than 5, while preserving the covering property. By fact 2.3, each remaining pair is axis-aligned at an axis-distance $\delta \in \{1, \dots, 5\}$.

Fix a lattice line ℓ . A pair at positions $(p, p+\delta)$ is contained in the windows whose integer starts satisfy

$$s \in [p + \delta - 5, p].$$

There are exactly $6 - \delta \leq 5$ such starts. If the left cells of consecutive pairs are p_i and p_{i+1} , coverage of every start requires $p_{i+1} - p_i \leq 6 - \delta_{i+1} \leq 5$. Allowing mixed values of δ cannot improve this bound. A segment of ℓ with L cells has $L - 5$ internal window starts and therefore contains at least $(L - 5)/5$ pairs. At least $2(L - 5)/5$ of its cells are matched along the axis of ℓ .

Each matched cell serves exactly one axis, namely the axis of its pair. Let

$$B_R = \{x : d(x, 0) \leq R\}, \quad |B_R| = 3R^2 + 3R + 1.$$

For each of the three axes, B_R meets exactly $2R+1$ line segments, and the lengths of those segments sum to $|B_R|$. Summing the line bound over all segments and all three axes gives

$$\text{matched-cell incidences} \geq \frac{6}{5}|B_R| - 6(2R + 1).$$

A matching supplies at most $|B_R|$ incidences. At $R = 20$, however,

$$\frac{6}{5}(1261) - 246 = 1267.2 > 1261,$$

which is a contradiction. $\square$

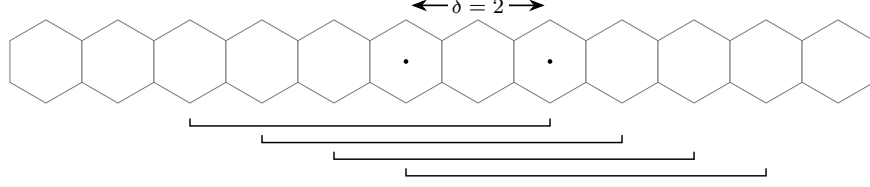


Figure 7: A pair at axis-distance $\delta = 2$ is contained in exactly $6 - \delta = 4$ length-six windows. The four brackets show their possible starts.

The same density calculation applies to other translation-invariant line systems with negligible ball boundary.

Corollary 7.2 (Translation-invariant line grids; source T8.1). *Consider k -in-a-row on a point lattice with g pairwise nonparallel, translation-invariant line directions. Suppose every lattice line has a length- k window at every offset and the lattice balls have boundary-to-volume ratio tending to zero, as in the Følner property of $\mathbb{Z}^2$. A pairing requires per-line matched density $2/(k-1)$ and hence*

$$g \frac{2}{k-1} \leq 1 \quad \implies \quad k \geq 2g + 1.$$

For the square grid, $g = 4$ gives $k \geq 9$, consistent with the classical existence of pairing draws for 9-in-a-row. For the hex grid, $g = 3$ gives $k \geq 7$, so 6-in-a-row misses the bound by one. Existence at the threshold requires a separate construction and is not claimed. The conclusion applies to lattices satisfying the stated hypotheses, not to arbitrary grids.

The global-pairing approach is therefore closed. Partial pairings on constrained regions are not ruled out, but they cannot be position-independent.

8 Potential-function certificates

The potential method follows Erdős and Selfridge's hypergraph criterion and its biased-game form developed by Beck [3, 4]. Here it supplies sufficient blocking certificates for selected Hexo windows. Window births prevent the direct global invariant from giving an unrestricted forever theorem.

8.1 Potential and the round inequality

The potential counts only attacker-touched windows that remain available to the attacker.

Definition 8.1 (Residual potential; companion D3 and D4). Let F be a finite labelled family of attacker target windows. At a position P , write A and D for the sets of cells occupied by attacker and defender stones, respectively, and put

$$\mathcal{L}_F(P) = \{W \in F : W \cap D = \emptyset\}, \quad E_P(W) = W \setminus A,$$

and set

$$\lambda = \sqrt{3}, \quad \Psi_F(P) = \sum_{W \in \mathcal{L}_F(P)} \lambda^{-|E_P(W)|}.$$

The dynamic potential is

$$\Phi(P) = \sum_{\substack{W: W \cap A \neq \emptyset \\ W \cap D = \emptyset}} \lambda^{-e_P(W)}, \quad e_P(W) = |W \setminus A|.$$

Thus Φ includes exactly the attacker-touched alive windows. All-empty windows are excluded: including them would add $\lambda^{-6} = 1/27$ for infinitely many windows and make the sum diverge.

For an empty cell x , define its fixed-family danger by

$$d_F(x; P) = \sum_{\substack{W \in \mathcal{L}_F(P) \\ x \in E_P(W)}} \lambda^{-|E_P(W)|}.$$

Sequential greedy defense recomputes this danger before each of the defender's two placements and chooses a maximum-danger empty cell, using fixed deterministic tie-breaking. When every scored danger is zero, the filler chooses an occupied cell of maximum q , breaking ties by maximum r , and places at its neighbor one step in the positive Q direction.

The value $\lambda = \sqrt{3}$ is the largest, and hence optimal, base supported by the two-defender, two-attacker round calculation.

Lemma 8.2 (One full fixed-family round; companion L5). *Let the defender make two sequential greedy placements in a fixed residual family, followed by at most two attacker placements. Assume either that all positive-danger cells are legal or that every surviving label is attacker-alive. The fixed-family potential at the next defender epoch is no larger than at the preceding defender epoch.*

Proof. Let δ_1 and δ_2 be the exact reductions made by the two greedy defender placements. After both placements, let

$$S = \max_x d_F(x)$$

over the remaining empty cells. Deleting edges never increases a cell's danger. The first and second greedy maxima are therefore both at least S , so

$$\delta_1 + \delta_2 \geq 2S. \tag{1}$$

An attacker placement at x multiplies by λ the weight of each residual edge containing x , and hence increases the potential by exactly $(\lambda-1)d_F(x)$. After one attacker placement every remaining danger is at most λS . The two attacker placements therefore increase the potential by at most

$$(\lambda-1)(1+\lambda)S = (\lambda^2-1)S = 2S. \tag{2}$$

Equations (1)–(2) show that the full round change is nonpositive. If play stops after the first attacker placement, omitting the second nonnegative increase only strengthens the inequality up to that placement. $\square$

Base 2 does not support the claimed blocking threshold. Take three target windows so far apart that no length-six window intersects two of them, and put four attacker stones in each target. Each target then has weight 2^{-2} , so the sum is $3/4 < 1$. The defender can kill at most two separated targets in one turn. The attacker fills the two empties of the survivor and wins.

8.2 Certificates and their boundary

The fixed-family theorem applies only at the start of a full defender turn.

Theorem 8.3 (Fixed-family Hexo blocking; companion Theorem 1). *Let P_0 be a nonterminal defender-FirstStone position, and let F be any finite fixed family of windows that are attacker-alive at P_0 . If*

$$\Psi_F(P_0) < 1,$$

then sequential greedy defense prevents the attacker from ever completing a window in F .

Proof in the companion. The round inequality keeps $\Psi_F < 1$ at every defender epoch and through the relevant prefixes of the following attacker pair, whereas a completed target has a term of weight 1. The full-turn premise is used for both greedy reductions. With only one defender placement remaining, two separated count-4 targets have potential $2/3 < 1$, but one survives and is completed. The condition is sufficient only; one defender stone may kill as many as 18 windows, so necessity is false. The complete proof is in [5]. $\square$

For the dynamic family, newly touched windows must be enrolled after they are born.

Definition 8.4 (Defender epochs and birth mass; companion D5). Let P_t be the position at the start of the defender's t th full turn. During the following attacker pair, or its prefix before a terminal placement, let N_t consist of the windows that were stone-free at P_t , were not hit by the intervening defender pair, and first became attacker-alive during that attacker pair. For $W \in N_t$, let $a_t(W) \in \{1, 2\}$ be the number of stones from that pair in W at the end of the pair or terminal prefix. Its exact birth mass is

$$C_t = \sum_{W \in N_t} \lambda^{-(6-a_t(W))}.$$

After a terminal play, all later source terms are zero.

The potential below one does certify the next complete attack cycle.

Theorem 8.5 (Global one-cycle certificate; companion Theorem 2). *If the defender is at FirstStone and the current dynamic potential satisfies $\Phi(P_t) < 1$, then two sequential touched-window greedy placements prevent the attacker from winning during the immediately following two-placement turn.*

Proof in the companion. Theorem 8.3 blocks every currently alive window, including a completion on the first attacker placement. A different potential winning window either already has a permanent defender stone or was stone-free at P_t and cannot receive six stones in one attacker pair. This is not a forever certificate, because births may put $\Phi(P_{t+1})$ above 1. The complete proof is in [5]. $\square$

A forever statement requires a bound that is uniform over attacker strategies, not a source sum observed along one path.

Theorem 8.6 (Cumulative-birth forever certificate; companion Theorem 3). *Fix the tie-breaking and filler policy in Definition 8.1, and use sequential touched-window greedy defense from P_0 . Suppose a uniform constant B_∞ satisfies*

$$\sup_{\tau} \sum_{t=0}^{\infty} C_t(\tau) \leq B_\infty, \quad \Phi(P_0) + B_\infty < 1,$$

where the supremum is over every legal attacker strategy τ . Then the defender prevents the attacker from ever completing a window. More generally, if B_T is a uniform upper bound satisfying

$$\sup_{\tau} \sum_{t=0}^{T-1} C_t(\tau) \leq B_T, \quad \Phi(P_0) + B_T < 1,$$

then the first T attacker pairs, indexed $0, \dots, T-1$, are certified.

Proof in the companion. The dynamic recurrence $\Phi(P_{t+1}) \leq \Phi(P_t) + C_t$ and induction keep the potential below one before each certified pair, so [Theorem 8.5](#) applies repeatedly. The supremum is used to make the same argument against every attacker strategy; a pathwise observed value of $\sum_t C_t$ has the wrong quantifier and is not a certificate. The complete proof is in [5]. $\square$

Finite confinement supplies a computable reservoir of virgin windows.

Definition 8.7 (Finite-region targets; companion D6). Fix a defender epoch P_0 and a finite set R of cells empty at P_0 . Assume every future attacker placement is confined to R . Define

$$\begin{aligned}\mathcal{A}_R &= \{W : W \cap D_0 = \emptyset, W \cap A_0 \neq \emptyset, W \setminus A_0 \subseteq R\}, \\ \mathcal{V}_R &= \{W : W \text{ is stone-free at } P_0, W \subseteq R\}, \quad N_R = |\mathcal{V}_R|,\end{aligned}$$

and

$$\Phi_R(P_0) = \sum_{W \in \mathcal{A}_R} \lambda^{-e_{P_0}(W)}.$$

Region-target greedy scores only currently attacker-alive members of the fixed target family $\mathcal{H}_R = \mathcal{A}_R \cup \mathcal{V}_R$.

Definition 8.8 (Virgin activation reserve; companion D7 and Lemma 9). For a matching M of pairwise vertex-disjoint two-cell subsets of R , let

$$q(M) = |\{W \in \mathcal{V}_R : \text{some } \{x, y\} \in M \text{ satisfies } x, y \in W\}|,$$

and put

$$P_R = \max_M q(M), \quad B_R^* = N_R \lambda^{-5} + P_R(\lambda^{-4} - \lambda^{-5}).$$

For certification, P_R must be evaluated exactly or replaced by a proved upper bound, such as $P_R \leq \min\{N_R, 5\lfloor |R|/2 \rfloor\}$. An exhibited matching gives only a lower bound and is unsafe here. The coarser reserve is $B_R^* \leq N_R/9$.

Theorem 8.9 (Finite-region forever theorem; companion Theorem 4). *If the attacker is forever confined to the finite region R and*

$$\Phi_R(P_0) + B_R^* < 1,$$

then region-target greedy defense prevents the attacker from ever winning. The weaker sufficient condition

$$\Phi_R(P_0) + \frac{N_R}{9} < 1$$

has the same conclusion. Every positive-reduction defender placement made by this strategy is an empty cell of a currently attacker-alive member of $\mathcal{H}_R$ and is legal. The confinement is on the attacker; defender fillers may lie outside R .

Proof in the companion. Delayed enrollment gives the fixed-target recurrence, while the matching reserve charges each virgin window once and charges the larger two-stone birth weight to at most P_R windows. Every window completable under confinement belongs to $\mathcal{A}_R$ or $\mathcal{V}_R$. The confinement premise must come from a separate certificate; the theorem does not establish it. The complete proof is in [5]. $\square$

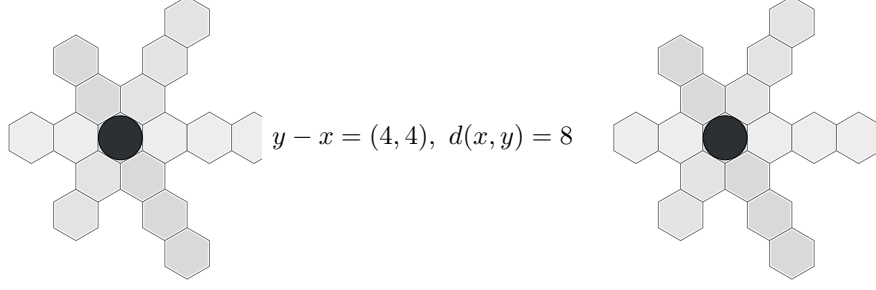


Figure 8: Counterexample 1, schematically. The two attacker placements are at hex distance 8, off all three axes, and have disjoint clean window stars. Together they enroll 36 one-stone windows, and the potential jumps from 0 to $36\lambda^{-5} = 4/\sqrt{3} > 1$. Three of the 18 windows through each new stone are shaded.

8.3 Practical domain and open boundary

The exact node test can be performed without floating-point arithmetic.

Corollary 8.10 (Integer-only threshold check; companion Corollary 2). *At a nonterminal position, for $s = 1, \dots, 5$, let n_s be the number of currently attacker-alive windows with exactly s attacker stones. Put*

$$a = n_1 + 3n_3 + 9n_5, \quad b = n_2 + 3n_4.$$

Then

$$\Phi < 1 \iff b \leq 8, \quad a^2 < 3(9 - b)^2.$$

Indeed, $\Phi < 1$ is $a + \sqrt{3}b < 9\sqrt{3}$. The right side of $a < \sqrt{3}(9 - b)$ must be positive, giving $b \leq 8$, after which squaring is equivalent. This check costs $O(\#\text{touched windows})$, not $O(\#\text{threats})$.

If $I = \sum_{s=1}^5 sn_s$ is the number of attacker-stone/alive-window incidences, then $\Phi \geq I/18$. Thus $\Phi < 1$ forces $I \leq 17$. With N_A attacker stones, all but at most 17 of the $18N_A$ incidences must already lie in defender-killed windows. The practical domain is therefore a heavily blocked, endgame-like region rather than a general middlegame.

The raw global claim that $\Phi < 1$ blocks the unrestricted infinite game forever remains open. Counterexample 1 starts with $\Phi = 0$ and, after the fixed zero-danger fillers, uses two far placements to birth 36 one-stone windows, producing $4/\sqrt{3} > 1$ as shown in Fig. 8. Requiring that no current attacker window have at most two empties does not repair the invariant, because the condition holds vacuously at $\Phi = 0$ before the same births. Replacing $\Phi < 1$ by $\Phi \leq 1$ also fails: three separated count-4 targets have total potential 1, the defender kills at most two, and the attacker completes the survivor. These are failures of the proposed potential certificates, not proofs that the attacker wins every corresponding unrestricted game. The $\Phi = 1$ example is a static blanket-game construction; it is not asserted to be a material-balanced reachable engine position and does not prove sharpness after restriction to reachable nodes.

Every positive-reduction greedy placement lies in an attacker-touched alive window and therefore inside the relevance closure of Section 6. If no monitored window remains, one or both placements of a defender turn may instead be arbitrary legal fillers. The certified strategy must permit these fillers. Their presence does not affect the dismissal soundness theorem; it only constrains the potential-certified strategy.

9 Domination patterns

Domination compares a searched defender reply with an omitted reply over a finite stopped horizon. The comparison transfers complete strategies and checks occupancy, window masks, and the legality frontier after every mapped placement. This prevents local mask similarity from hiding a different legal continuation.

For a cell x , write $B_R(x) = \{z : d(x, z) \leq R\}$ and let $\Omega(x)$ be the 18 windows containing x . For a post-opening position P , define its geometric support by

$$\Lambda(P) = \bigcup_{s \in \text{Stones}(P)} B_8(s).$$

If P is nonterminal, its legal placements are $L(P) = \Lambda(P) \setminus \text{Stones}(P)$; if P is terminal, $L(P) = \emptyset$. An empty cell is *dead* when every window in $\Omega(x)$ is dead.

9.1 Stopped outcomes and causal simulation

The outcome records the first terminal placement, with the defender's reply itself at time zero.

Definition 9.1 (Stopped horizon outcome; companion D3 and D4). Let S be a possibly terminal successor of a defender reply. A horizon n allows at most n further placements after that reply. Given pure continuation strategies σ_D and τ_A , play stops at the first terminal placement or after those n placements, and

$$\text{Out}_n(S, \sigma_D, \tau_A) \in \{D@t : 0 \leq t \leq n\} \cup \{A@t : 0 \leq t \leq n\} \cup \{?\}.$$

Here $?$ means that no terminal placement occurred through the horizon. For defender comparison, set

$$u_D(D@t) = 1, \quad u_D(?) = 0, \quad u_D(A@t) = -1.$$

Winning windows and times remain in the trace, although all wins by one player have the same qualitative value. A pure X -strategy assigns a legal placement to every nonterminal history of length less than n at which X is to place, including both consecutive histories of an X turn, and is never queried after a terminal history. Write $\Sigma_X(S)$ for the set of such strategies from S .

The transfer directions encode the defender-pruning implication.

Definition 9.2 (n -outcome domination; companion D5 and Lemma 3). Let a and b be legal defender replies at the same nonterminal position P , with stopped successors $S_a = P + a$ and $S_b = P + b$. The reply b is *n -outcome-dominated by a for the defender*, written $b \preceq_n a$, if there are maps

$$T_D : \Sigma_D(S_b) \rightarrow \Sigma_D(S_a), \quad T_A : \Sigma_D(S_b) \times \Sigma_A(S_a) \rightarrow \Sigma_A(S_b)$$

such that, for every discarded-branch defender strategy σ_b and every searched-branch attacker strategy τ_a ,

$$u_D(\text{Out}_n(S_a, T_D(\sigma_b), \tau_a)) \geq u_D(\text{Out}_n(S_b, \sigma_b, T_A(\sigma_b, \tau_a))).$$

Equivalently,

$$\forall \sigma_b \exists \sigma_a \forall \tau_a \exists \tau_b : \quad u_D(\text{Out}_n(S_a, \sigma_a, \tau_a)) \geq u_D(\text{Out}_n(S_b, \sigma_b, \tau_b)).$$

Consequently, an attacker win forced by the horizon after the searched reply a is also forced after the omitted reply b .

A causal simulation is the checkable witness used by the patterns below.

Definition 9.3 (Causal alternating simulation; companion D6). A depth- k defender-favouring alternating simulation relates a searched state G to an omitted state H at equal placement depth. A defender-terminal G , or an attacker-terminal H , is accepted immediately. An attacker-terminal G is accepted only if H is already attacker-terminal; a defender-terminal H is accepted only if G is already defender-terminal. No move is generated after a terminal state. Two nonterminal states are accepted when $k = 0$.

Otherwise the states have the same player to move and the same remaining placement budget. At a defender history, every legal move in H is mapped causally to a legal move in G . At an attacker history, every legal move in G is mapped causally to a legal move in H . The successor pair is related at depth $k - 1$. The map depends only on the paired prefix and current move. It pairs FirstStone with FirstStone and SecondStone with SecondStone, including stored first-cell coordinates under any coordinate involution.

Such a simulation implies n -outcome domination by recursively transferring the two players' strategies [5]. Each simulation step must account for three channels: the mapped cells must be empty with explicit images for branch-exclusive empties; after each mapped placement, every labelled mask in the union of the two incident 18-window sets must be compared, with the terminal clauses applied before phase advance; and legal moves must be included in the actor-dependent direction.

9.2 Dead cells and interchangeable hits

A dead empty has no legality frontier because the dead-window witnesses on the six directed rays already support its whole radius-eight ball.

Lemma 9.4 (A dead empty has no legality frontier; companion Lemma 7). *Let b be empty and dead in a nonterminal post-opening position P . Then*

$$B_8(b) \subseteq \Lambda(P).$$

In particular, b is legal, and placing at b inserts no previously unsupported cell into the legal store.

Proof. Let u range over the six directed axial unit vectors. The endpoint window

$$W_u = \{b, b + u, \dots, b + 5u\}$$

belongs to $\Omega(b)$. It is dead while b is empty, so among $b + u, \dots, b + 5u$ there is an old stone $s_u = b + k_u u$ with $1 \leq k_u \leq 5$.

Take $z \in B_8(b)$. It lies in a closed 60° sector between adjacent directed axes u and v , so

$$z - b = \alpha u + \beta v, \quad \alpha, \beta \geq 0, \quad \alpha + \beta = d(b, z) \leq 8.$$

After a rotation or reflection, take $u = (1, 0)$ and $v = (0, 1)$, and write $s = s_u = b + ku$ with $1 \leq k \leq 5$. Then

$$d(z, s) = \max\{|\alpha - k|, \beta, |\alpha + \beta - k|\} \leq 8,$$

because $\alpha, \beta, \alpha + \beta$, and k all lie in $[0, 8]$. Thus $z \in B_8(s) \subseteq \Lambda(P)$. This holds for every z . $\square$

Deadness makes the omitted reply frontier-inert, but it does not impose the same condition on the searched substitute.

Proposition 9.5 (Frontier-certified dead-cell dismissal; companion P1). *Let P be a nonterminal post-opening position with the defender to place. Let $a \neq b$ be legal empty cells. Assume every window in $\Omega(b)$ is already dead in P . Unless $P + a$ is an immediate defender win, also assume*

$$\Lambda(P) \cup B_8(a) = \Lambda(P) \cup B_8(b).$$

Then $b \preceq_n a$ for every finite n . By Lemma 9.4, the displayed condition is equivalent here to $B_8(a) \subseteq \Lambda(P)$. Thus b may be dismissed in favor of any searched a that either wins immediately or is frontier-inert. Deadness of b does not make this separate obligation on a automatic.

Proof idea; complete proof in the companion. Compare $P + a$ and $P + b$ by transposing a and b until the counterpart is used. The hypotheses synchronize legal support. Every window through b is permanently dead, while the remaining changed windows either agree or change ownership in the defender-favouring direction. The occupancy, terminality, and frontier channels therefore give an all-depth causal simulation; an immediate defender win after a is already maximal. See [5]. $\square$

Two hits of one count-4 threat become symmetric only after every other incident window is made inert and the successor supports agree.

Proposition 9.6 (Dead-spoke interchangeable hits; companion P2). *Let P be a nonterminal post-opening position with the defender to place. Let W be an attacker-alive window with exactly four attacker stones and exactly two empty cells $x \neq y$. Both x and y are legal. Assume*

$$(\Omega(x) \cup \Omega(y)) \setminus \{W\}$$

consists entirely of windows already dead in P , and assume

$$\Lambda(P) \cup B_8(x) = \Lambda(P) \cup B_8(y).$$

Then the replies are mutually outcome-dominating at every horizon:

$$x \preceq_n y, \quad y \preceq_n x \quad (n < \infty).$$

A solver may retain one canonical member of $\{x, y\}$ at that defender node.

Proof idea; complete proof in the companion. Either defender hit kills W . The dead-spoke premise then makes every window through either special cell permanently dead. Transposing x and y preserves occupancy choices, the support equality preserves legal moves in both directions, and every mask outside the dead union is identical. This is an outcome-preserving bisimulation at all depths. See [5]. $\square$

Equality of count profiles is insufficient. Labelled empty cells and the window-intersection hypergraph can change minimum hitting sets or fork-creating cells even when every pair $(\text{cnt}_A, \text{cnt}_D)$ agrees. A broader equivalence would have to preserve labelled masks, incidence, occupancy, and radius-eight support throughout the n -causal cone. The dead-spoke premise avoids that unproved classification.

9.3 Same-turn commutation

Two placements may be sorted only when both were legal before the turn and neither possible first placement wins.

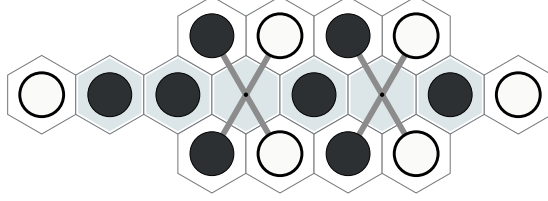


Figure 9: A representative dead-spoke configuration. The highlighted threat has four attacker stones and the two marked empty hits. Three spoke directions per hit show old attacker and defender witnesses. The P2 premise applies this deadness condition to all 17 other windows through each marked cell, not only the representative spokes shown.

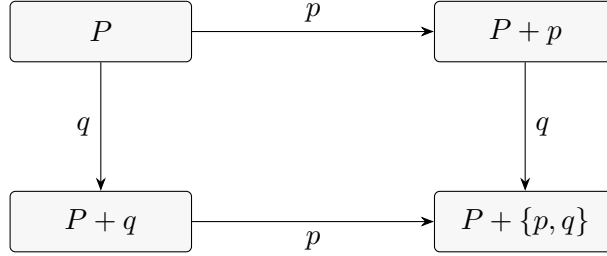


Figure 10: The commuting square for two same-turn placements. Both paths reach the same completed-turn rule state under the premises of P3.

Proposition 9.7 (Nonwinning-prefix turn commutation; companion P3). *Let P be a nonterminal FirstStone position for either player X . Let $p \neq q$ be legal already in P , and suppose neither $P + p$ nor $P + q$ is terminal. Then both traces $p; q$ and $q; p$ are legal and have the same qualitative rule outcome after the second placement. If nonterminal, they have identical occupancy, labelled window masks, legal support, player to move, and FirstStone phase. If terminal, both terminate on the second placement with winner X and the same absolute placement count.*

Proof idea; complete proof in the companion. Initial legality makes either second cell remain legal, and the final support $\Lambda(P) \cup B_8(p) \cup B_8(q)$ is order-independent. Inserting the two owner bits into every window mask commutes, including in a window containing both cells. The singleton-nonterminal premises exclude a first-placement win; any joint win is therefore completed by the second placement in both orders. See [5]. $\square$

The live engine states need not be field-identical. Ordered history, last-turn metadata, serialized order, and the retained terminal SecondStone first-cell payload may differ. The proposition equates rule outcomes, not these order-sensitive features or the two intermediate SecondStone threat states.

The analytic proofs have randomized local checks, not exhaustive machine proofs. MV-P3 tested 2,997 random legal same-mover pairs under the singleton-nonwin filter and found 0 commutation mismatches. MV-L7 tested 400 adversarial dead-cell configurations and found 0 legality-frontier violations; the worst minimum distance from a cell of the radius-eight ball to an old stone was 6, leaving margin 2. Full exhaustive quotient enumeration remains open.

These results do not claim that an arbitrary frontier-extending reply can substitute for a dead cell, that equal window counts make hitting cells interchangeable, or that stones of one colour are globally monotone-helpful when legal frontiers differ; no transferred strategy continues after a terminal placement, and the prospective exhaustive enumerations are specifications rather than executed evidence.

10 Mechanized validation

The validation harness uses a mini-model solver that was checked move-for-move against the *hexo_engine* implementation. These runs tested the predecessor closure as a search generator.

Two closure experiments were executed. First, the closure-restricted solver did not claim the engineered junction position. The junction cell was in the closure, and the defender survived after 1,396 uncapped nodes. Second, a bounded random check covered 91 attackable positions. The closure-restricted solver made 52 WIN claims, with no divergence from the full legal move set at the matched horizon; the run took 139 seconds. As a negative control, the earlier radius-2 experiments used the same harness and produced false WIN claims for unsound restrictions.

These experiments are search-behavior evidence for the predecessor heuristic. They do not validate the revised ranked closure in [Theorem 6.7](#) and are not part of the proofs of [Theorems 5.1, 6.1 and 6.7](#). The separate machine-verification status of the domination patterns is summarized in [Section 9](#).

11 Open problems

Sharpened budget. The local budget B and exposure E^D still count every defender placement before the relevant local resolution or exposure stop. A sharper debit would combine quiet placements F with a per-window forced-hit capacity H_W . This requires branchwise worst-case bookkeeping. No theorem here proves that compulsory hits are unavailable for filling a selected window, so the problem remains open.

Resolved frontier accounting. The role radius $8(r - 1)$, uniformly $8(B - 1)$, and the touched-virgin split resolve the earlier band-accounting problem. The virgin radius $8(E^D - 6)$ is sharp at exposure 7.

Pairing at the threshold. The density bound in [Corollary 7.2](#) leaves $k = 7$ and $g = 3$ open, where the required matched density is exactly one. This threshold case does not affect Hexo, whose winning length is $k = 6$. Existence at the threshold remains unsettled.

Formalization. The model and results through [Theorem 6.12](#) are finite combinatorics and are feasible to formalize in Lean or Coq. Such a formalization would provide a reference checker for [Items \(Z2\) and \(Z4\)](#), [clause \(Z5'\)](#), the B , E^D , and r inequalities, transition-inclusive envelopes, and DAG label and clock consistency.

A Notation reference

[Table 1](#) collects the notation of the main text. [Tables 2 and 3](#) give the crosswalk from paper numbering to the source tags and verifier labels of the repository proof document.

Table 1: Notation used in the main text.

Symbol	Meaning	Defining location
$d(x, y)$	Hex distance between cells x and y .	Definition 2.1
$\mathbf{a}$	An axis vector in $\{(1, 0), (0, 1), (1, -1)\}$.	Definition 2.2
W	A six-cell window along one axis.	Definition 2.2
$\text{cnt}_X(W, P)$	Number of stones of player X in W at P .	Definition 2.4
alive, dead, threat	Window states determined by the two colour counts.	Definition 2.6
$\text{own_win_now}(P)$	Current-turn completion predicate for the mover.	Definition 2.7
$\text{hit}(P)$	Multiset of empty-cell sets of opponent threat windows.	Definition 2.7
$\text{mhs}(P)$	Minimum size of a set meeting every member of $\text{hit}(P)$.	Definition 2.7
ply	One placement, indexed absolutely along a play.	Definition 2.8
$\mathfrak{D}(P, T)$	Defender placements strictly before horizon ply T .	Definition 2.8
$B(N)$	Admissible local bound on defender placements before resolution.	Definition 4.7
$\Omega(N)$	Live obligation roles at reachable descendants of N .	Definition 4.8
$\text{Prot}(N)$	Cells carrying live attacker-placement or leaf-witness-empty roles.	Definition 4.8
$r_N(y)$	Maximum live role rank carried by cell y .	Definition 4.9
$E_N^D(W)$	Defender exposure before the attacker wins or enters W .	Definition 4.10
$S(N)$	Searched legal defender replies at AND node N .	Definition 4.6
$\text{Legal}(P)$	Legal empty placement cells in position P .	Definition 2.5
$\text{Stones}(P)$	Occupied cells in position P .	Definition 2.4
$B_r(U)$	Cells within distance r of some cell of U .	Definition 4.12
$Z_{\text{dir}}, Z_{\text{seed}}, Z_{\text{touch}}, Z_{\text{virgin}}$	Four components of the ranked verifier zone.	Definition 4.12
$\text{hitting}(P)$	Union of the empty sets of attacker-threat windows.	Definition 6.6
$R_{\text{cert}}, R_{\text{search}}$	Mandatory ranked closure and heuristic search superset.	Definition 6.6
$K_b(N)$	Kernel defined by residual transversal number.	Theorem 6.5
X	Real-only dismissed defender stones in the coupling.	Definition 4.14
Y	Ghost-only searched filler defender stones in the coupling.	Definition 4.14
$\hat{X}$	History of every cell ever added to X .	Definition 4.14
$\Phi(P)$	Dynamic potential of attacker-touched alive windows.	Definition 8.1
λ	Potential base $\sqrt{3}$.	Definition 8.1
$\Psi_F(P)$	Residual potential for a fixed target family F .	Definition 8.1

Table 2: Crosswalk from paper numbering to source definitions, facts, and lemmas in the repository proof document.

Paper number	Source tag	Name
Definition 2.1	D1	Board, cells, and distance
Definition 2.2	D2	Axes and windows
Definition 2.4	D3	Positions
Definition 2.5	D4	Legality and transitions
Definition 2.6	D5	Window states
Definition 2.7	D6	λ^1 predicates
Definition 2.8	D7	Plies and horizon
Definition 2.9	D8	Dominance direction
Definition 4.6	D9	Certificate grammar
Definition 4.8	D10	Compressed obligations
Definition 4.12	D11	Zone-carrying certificate
Definition 4.14	D12	Real-ghost coupling
Definition 6.6	D13	Ranked relevance closure
Definition 4.7	D14	Admissible local defender budget
Definition 4.9	D15	Cell-specific deadlines
Definition 4.10	D16	Per-window defender exposure
Definition 6.8	D17	Transition-inclusive substitution envelope
Definition 6.11	D18	Finite acyclic certificate DAG
Fact 2.3	F1	Window collinearity
Lemma 3.1	L1	Packing bound
Lemma 3.2	L2	Boundary pair
Lemma 4.1	L3	Difference accounting
Lemma 4.2	L4	Permanence
Lemma 4.3	L5	Inert-extension transfer
Lemma 4.4	L6	Deletion monotonicity and well-formed legality
Lemma 4.5	L7	Completion counting
Lemma 4.13	L9'	First protected occupation
Lemma 6.2	L10	Short-range attacker-obligation containment
Lemma 4.11	L11	Local-clock and nesting facts
Lemma 4.15	L12	Split completion safety
Lemma 6.3	L13	Sparse LOSS witnesses
Lemma 6.10	L14	Internal own-win diagnostic

Table 3: Crosswalk from paper numbering to source theorems and verifier labels in the repository proof document.

Paper number	Source tag	Name
Theorem 3.3	T1	Single-turn tactical locality
Theorem 3.4	T2	Threat-creation locality
Theorem 5.1	T3	Local-clock, pathwise dismissal soundness
Theorem 6.1	T4	Ranked zone reading
Theorem 6.4	T5	Static-set coverage for short local budgets
Theorem 6.5	T6	Extendable-hit kernel
Theorem 6.7	T7	Ranked-closure coverage
Theorem 7.1	T8	No pairing
Corollary 7.2	T8.1	Translation-invariant line grids
Theorem 6.9	T9	Branch-indexed substitution soundness
Theorem 6.12	T10	DAG unfolding
Removed in v2; remark in Section 4	Z1	Hitting hypothesis
Item (Z2)	Z2	Direct and completion protection
Item (Z4)	Z4	Well-formed attacker legality
Clause (Z5')	Z5'	Ranked obligation seed guard
Equation (MI)	MI	Anchored defender-count inequality

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